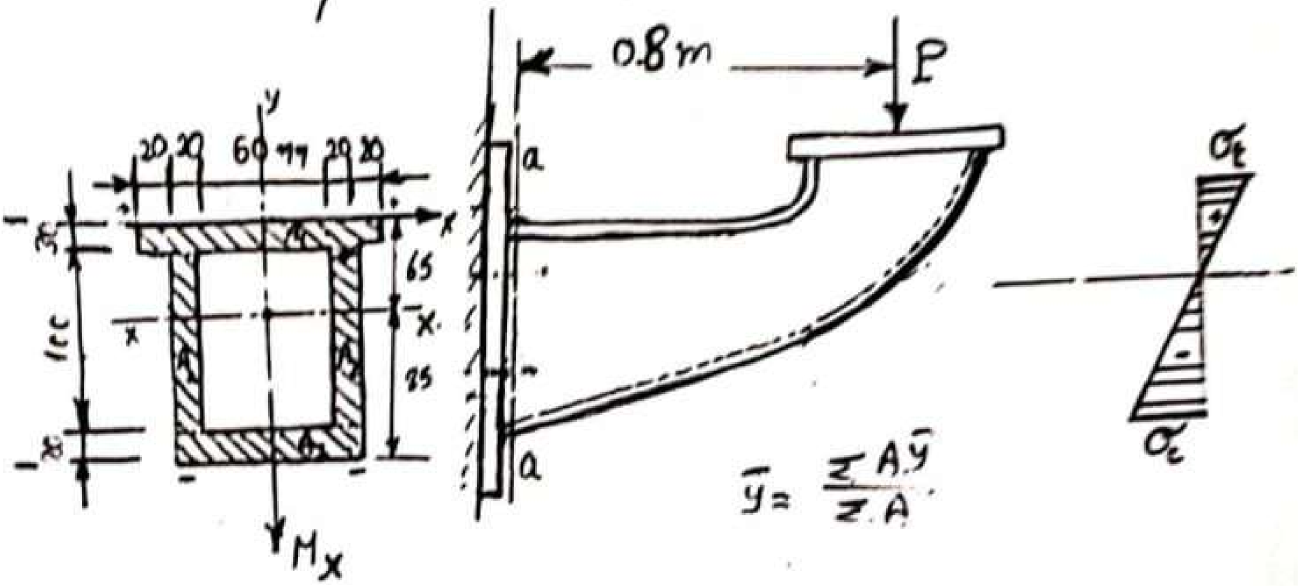
Determine the maximum value of the load P that may be applied to the machine bracket shown in the figure so that the extreme stresses do not exceed 40 MPa in tenson and 80 MPa in compression. (neglect effect of transverse shear).



$$\bar{y} = \frac{\Sigma A\bar{y}}{\Sigma A}$$

$$\bar{y} = \frac{30\times140\times15 + 20\times100\times80\times2 + 100\times20\times140}{30\times140 + 20\times100\times2 + 100\times20} = \underline{65 \text{ mm}}$$

$$M_x = P\times0{,}8 = 0{,}8\,P \quad \text{N.m}$$

$$I_x = \frac{140\times30^3}{12} + 140\times30(50)^2 + 2\left[\frac{20(100)^3}{12} + 20\times100(15)^2\right] + \frac{100(20)^3}{12} + 20\times100(75)^2$$

$$I_x = 264\times10^6 \; [\text{mm}^4] =$$

$$\sigma_t = \frac{M_x}{I_x}y_1 \le \sigma_{t_{all}} \qquad 40\times10^6 (\text{Pa}) \ge \frac{0{,}8\,P(\text{N.m})\times0{,}065(\text{m})}{26.4\times10^6\times10^{-12}(\text{m}^4)}$$

$$P \le \underline{20.3\times10^3 \text{ N}}$$

$$\sigma_c = \frac{M_x}{I_x}y_2 \le \sigma_{c_{all}} \qquad 80\times10^6 (\text{Pa}) \ge \frac{0{,}8P(\text{N.m})\times0{,}085(\text{m})}{26.4\times10^{-6}(\text{m}^4)}$$

$$P \le \underline{31.05\times10^3 \text{ N}}$$

The safe load = 20.3 kN